ia_study2_computeMeasures

2026-03-12

Admin (Packages & Custom Functions)

```r
#setwd(dirname(rstudioapi::getActiveDocumentContext()$path))
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.6.0
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.2.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
source("https://raw.githubusercontent.com/vyqlua/avi_functions/refs/heads/main/ipsatize_fn.R")
```

Load Data

```r
data <- read_csv("ia_study2_anon_data.csv")
data <- data[ , -1]
```

Ten Item Personality Inventory (TIPI)

Next, we compute the composite scores of the participant-level TIPI variables. Specifically, we calculate:

1. participant's extraversion `tipi_ex`

- calculated by averaging `tipi_1` (Extraverted, enthusiastic), `tipi_6` (reversed; Reserved, quiet)

2. participant's agreeableness, `tipi_a`

- calculated by averaging `tipi_2` (reversed; Critical, quarrelsome), `tipi_7` (Sympathetic, warm)

3. participant's conscientiousness, `tipi_c`

- calculated by averaging `tipi_3` (Dependable, self-disciplined), `tipi_8` (reversed; Disorganized, careless)

4. participant's emotoinal stability, `tipi_es`

- calculated by averaging `tipi_4` (reversed; Anxious, easily upset), `tipi_9` (Calm, emotionally stable)

5. participant's openness to experience, `tipi_o`

- calculated by averaging `tipi_5` (Open to new experiences, complex), `tipi_10` (reversed; Conventional, uncreative)

```r
# compute tipi variables
data <- data %>%
  rowwise() %>%
  dplyr::mutate( # reverse code tipi items
    tipi_6r = 8 - tipi_6,
    tipi_2r = 8 - tipi_2,
    tipi_8r = 8 - tipi_8,
    tipi_4r = 8 - tipi_4,
    tipi_10r = 8 - tipi_10
    ) %>%
  dplyr::mutate( # calculate tipi_ex, tipi_a, tipi_c, tipi_es, and tipi_o
    tipi_ex = mean(c(tipi_1, tipi_6r), na.rm = T),
    tipi_a = mean(c(tipi_2r, tipi_7), na.rm = T),
    tipi_c = mean(c(tipi_3, tipi_8r), na.rm = T),
    tipi_es = mean(c(tipi_4r, tipi_9), na.rm = T),
    tipi_o = mean(c(tipi_5, tipi_10r), na.rm = T)
    ) %>%
  ungroup()
```

```r
data %>%
  dplyr::select("tipi_1", "tipi_6r",
                "tipi_2r", "tipi_7",
                "tipi_3", "tipi_8r",
                "tipi_4r", "tipi_9",
                "tipi_5", "tipi_10r") %>%
  cor(use = "pairwise.complete.obs") %>%
  round(2)
```

```
##          tipi_1 tipi_6r tipi_2r tipi_7 tipi_3 tipi_8r tipi_4r tipi_9 tipi_5
## tipi_1     1.00    0.52   -0.07   0.08   0.22    0.01    0.13   0.09   0.27
## tipi_6r    0.52    1.00    0.05   0.08  -0.02   -0.02    0.03  -0.17   0.09
## tipi_2r   -0.07    0.05    1.00   0.17   0.15    0.22    0.09   0.05  -0.03
## tipi_7     0.08    0.08    0.17   1.00   0.22    0.11   -0.04   0.10   0.10
## tipi_3     0.22   -0.02    0.15   0.22   1.00    0.46    0.12   0.23   0.07
## tipi_8r    0.01   -0.02    0.22   0.11   0.46    1.00    0.15   0.15  -0.03
## tipi_4r    0.13    0.03    0.09  -0.04   0.12    0.15    1.00   0.44   0.23
## tipi_9     0.09   -0.17    0.05   0.10   0.23    0.15    0.44   1.00   0.16
## tipi_5     0.27    0.09   -0.03   0.10   0.07   -0.03    0.23   0.16   1.00
## tipi_10r   0.24    0.17    0.10   0.04  -0.03    0.07    0.09  -0.13   0.31
##          tipi_10r
## tipi_1       0.24
## tipi_6r      0.17
## tipi_2r      0.10
## tipi_7       0.04
## tipi_3      -0.03
## tipi_8r      0.07
## tipi_4r      0.09
## tipi_9      -0.13
## tipi_5       0.31
## tipi_10r     1.00
```

```r
# European American
data %>% dplyr::filter(cultureEA == 1) %>%
  dplyr::select("tipi_1", "tipi_6r",
                "tipi_2r", "tipi_7",
                "tipi_3", "tipi_8r",
                "tipi_4r", "tipi_9",
               "tipi_5", "tipi_10r") %>%
  cor(use = "pairwise.complete.obs") %>%
  round(2)
```

```
##          tipi_1 tipi_6r tipi_2r tipi_7 tipi_3 tipi_8r tipi_4r tipi_9 tipi_5
## tipi_1     1.00    0.47   -0.07  -0.03   0.24    0.06    0.20   0.28   0.34
## tipi_6r    0.47    1.00    0.16  -0.03  -0.02    0.12    0.06  -0.05   0.10
## tipi_2r   -0.07    0.16    1.00   0.11   0.06    0.21    0.13  -0.05  -0.03
## tipi_7    -0.03   -0.03    0.11   1.00   0.30    0.18    0.00   0.14   0.17
## tipi_3     0.24   -0.02    0.06   0.30   1.00    0.39    0.10   0.35   0.13
## tipi_8r    0.06    0.12    0.21   0.18   0.39    1.00    0.03   0.14  -0.11
## tipi_4r    0.20    0.06    0.13   0.00   0.10    0.03    1.00   0.44   0.32
## tipi_9     0.28   -0.05   -0.05   0.14   0.35    0.14    0.44   1.00   0.21
## tipi_5     0.34    0.10   -0.03   0.17   0.13   -0.11    0.32   0.21   1.00
## tipi_10r   0.22    0.19    0.11   0.01  -0.04    0.06    0.09  -0.18   0.26
##          tipi_10r
## tipi_1       0.22
## tipi_6r      0.19
## tipi_2r      0.11
## tipi_7       0.01
## tipi_3      -0.04
## tipi_8r      0.06
## tipi_4r      0.09
## tipi_9      -0.18
## tipi_5       0.26
```

```
## tipi_10r     1.00
```

```r
# Taiwanese
data %>% dplyr::filter(cultureEA == 0) %>%
  dplyr::select("tipi_1", "tipi_6r",
                "tipi_2r", "tipi_7",
                "tipi_3", "tipi_8r",
                "tipi_4r", "tipi_9",
                "tipi_5", "tipi_10r")%>%
  cor(use = "pairwise.complete.obs") %>%
  round(2)
```

```
##          tipi_1 tipi_6r tipi_2r tipi_7 tipi_3 tipi_8r tipi_4r tipi_9 tipi_5
## tipi_1     1.00    0.61   -0.07   0.21   0.18   -0.11    0.04  -0.15   0.16
## tipi_6r    0.61    1.00   -0.08   0.25   0.06   -0.09    0.02  -0.29   0.10
## tipi_2r   -0.07   -0.08    1.00   0.26   0.26    0.27    0.05   0.17  -0.02
## tipi_7     0.21    0.25    0.26   1.00   0.08   -0.04   -0.11   0.05   0.00
## tipi_3     0.18    0.06    0.26   0.08   1.00    0.44    0.12   0.11  -0.02
## tipi_8r   -0.11   -0.09    0.27  -0.04   0.44    1.00    0.26   0.14   0.03
## tipi_4r    0.04    0.02    0.05  -0.11   0.12    0.26    1.00   0.43   0.11
## tipi_9    -0.15   -0.29    0.17   0.05   0.11    0.14    0.43   1.00   0.09
## tipi_5     0.16    0.10   -0.02   0.00  -0.02    0.03    0.11   0.09   1.00
## tipi_10r   0.26    0.19    0.10   0.05  -0.08    0.01    0.07  -0.09   0.37
##          tipi_10r
## tipi_1       0.26
## tipi_6r      0.19
## tipi_2r      0.10
## tipi_7       0.05
## tipi_3      -0.08
## tipi_8r      0.01
## tipi_4r      0.07
## tipi_9      -0.09
## tipi_5       0.37
## tipi_10r     1.00
```

Interpersonal Reativity Index (IRI)

Next, we compute the composite scores of the participant-level IRI variables. Specifically, we calculate:

1. participant's perspective-taking `iri_pt`

- calculated by averaging `iri3`, `iri8`, `iri11`, `iri15`, `iri21`, `iri25`, & `iri28`

2. participant's fantasy, `iri_fs`

- calculated by averaging `iri1`, `iri5`, `iri7`, `iri12`, `iri16`, `iri23`, & `iri26`

3. participant's empathetic concern, `iri_ec`

- calculated by averaging `iri2`, `iri4`, `iri9`, `iri14`, `iri18`, `iri20`, & `iri22`

4. participant's personal distress, `iri_pd`

- calculated by averaging `iri6`, `iri10`, `iri13`, `iri17`, `iri19`, `iri24`, & `iri27`

```
# compute iri variables
data <- data %>%
  rowwise() %>%
  dplyr::mutate( # reverse code iri items
    iri3r = 6 - iri3,
    iri4r = 6 - iri4,
    iri7r = 6 - iri7,
    iri12r = 6 - iri12,
    iri13r = 6 - iri13,
    iri14r = 6 - iri14,
    iri15r = 6 - iri15,
    iri18r = 6 - iri18,
    iri19r = 6 - iri19,
    ) %>%
  dplyr::mutate( # calculate iri_pt, iri_fs, iri_ec, iri_pd
    iri_pt = mean(
      c(iri3r, iri8, iri11, iri15r, iri21, iri25, iri28), na.rm = T),
    iri_fs = mean(
      c(iri1, iri5, iri7r, iri12r, iri16, iri23, iri26), na.rm = T),
    iri_ec = mean(
      c(iri2, iri4r, iri9, iri14r, iri18r, iri20, iri22), na.rm = T),
    iri_pd = mean(
      c(iri6, iri10, iri13r, iri17, iri19r, iri24, iri27), na.rm = T)
    ) %>%
  ungroup()
```

```
reliab_iri <- c("iri3r, iri8, iri11, iri15r, iri21, iri25, iri28",
               "iri1, iri5, iri7r, iri12r, iri16, iri23, iri26",
               "iri2, iri4r, iri9, iri14r, iri18r, iri20, iri22",
               "iri6, iri10, iri13r, iri17, iri19r, iri24, iri27")

reliab_iri_names <- c("iri_pt", "iri_fs", "iri_ec", "iri_pd")

all_alphas(data, items = reliab_iri, constructs = reliab_iri_names,
           sensitivity = sensitivity_analyses)
```

```
## # A tibble: 12 x 3
##    construct      `!is.na(subID)` `exclude_sd != 1`
##    <chr>                    <dbl>             <dbl>
##  1 iri_pt_overall            0.77              0.77
##  2 iri_fs_overall            0.74              0.71
##  3 iri_ec_overall            0.78              0.79
##  4 iri_pd_overall            0.77              0.76
##  5 iri_pt_EA                 0.8               0.81
##  6 iri_fs_EA                 0.75              0.72
##  7 iri_ec_EA                 0.75              0.77
##  8 iri_pd_EA                 0.75              0.72
##  9 iri_pt_TN                 0.73              0.7
## 10 iri_fs_TN                 0.72              0.71
```

```
## 11 iri_ec_TN                  0.78              0.78
## 12 iri_pd_TN                  0.79              0.8
```

AVI

Now, we compute the composite scores of the participant-level AVI variables.

```
avi_items <-  c(
  "1_10" = ".sleepy",
  "1_1"  = ".enthusiastic",
  "1_2"  = ".astonished",
  "1_3"  = ".nervous",
  "1_4"  = ".dull",
  "1_5"  = ".quiet",
  "1_6"  = ".relaxed",
  "1_7"  = ".excited",
  "1_8"  = ".surprised",
  "1_9"  = ".elated",
  "2_10" = ".fearful",
  "2_1"  = ".still",
  "2_2"  = ".lonely",
  "2_3"  = ".strong",
  "2_4"  = ".passive",
  "2_5"  = ".content",
  "2_6"  = ".sluggish",
  "2_7"  = ".inactive",
  "2_8"  = ".sad",
  "2_9"  = ".euphoric",
  "3_10" = ".serene",
  "3_1"  = ".happy",
  "3_2"  = ".idle",
  "3_3"  = ".calm",
  "3_4"  = ".unhappy",
  "3_5"  = ".aroused",
  "3_6"  = ".hostile",
  "3_7"  = ".satisfied",
  "3_8"  = ".rested",
  "3_9"  = ".peaceful",
  "4_1"  = ".noemotion",
  "4_2"  = ".energetic",
  "4_3"  = ".ashamed",
  "4_4"  = ".disgusted",
  "4_5"  = ".stressed",
  "4_6"  = ".guilty",
  "4_7"  = ".contemptuous",
  "4_8"  = ".fatigued",
  "4_9"  = ".angry")

names(data) <- names(data) |>
  str_replace(paste0(names(avi_items), collapse = "|"),
              avi_items[str_extract(names(data), paste(names(avi_items), collapse="|"))])
```

```
# actual affect
data <- ipsatize_avi(data,
                     item_stem = "r.",
                     remove = c("euphoric", "contemptuous", "angry"),
                     maximizing_pos = TRUE)
```

```
## Items ipsatized successfully. Variables created:

##  [1] "rMean"            "rSd"               "r.enthusiastic_i" "r.astonished_i"
##  [5] "r.nervous_i"      "r.dull_i"          "r.quiet_i"        "r.relaxed_i"
##  [9] "r.excited_i"      "r.surprised_i"     "r.elated_i"       "r.sleepy_i"
## [13] "r.still_i"        "r.lonely_i"        "r.strong_i"       "r.passive_i"
## [17] "r.content_i"      "r.sluggish_i"      "r.inactive_i"     "r.sad_i"
## [21] "r.euphoric_i"     "r.fearful_i"       "r.happy_i"        "r.idle_i"
## [25] "r.calm_i"         "r.unhappy_i"       "r.aroused_i"      "r.hostile_i"
## [29] "r.satisfied_i"    "r.rested_i"        "r.peaceful_i"     "r.serene_i"
## [33] "r.noemotion_i"    "r.energetic_i"     "r.ashamed_i"      "r.disgusted_i"
## [37] "r.stressed_i"     "r.guilty_i"        "r.contemptuous_i" "r.fatigued_i"
## [41] "r.angry_i"

## AVI Variables Computed:

##  [1] "rHAP"      "rLAP"      "rHAN"      "rLAN"      "rHAP_i"    "rLAP_i"
##  [7] "rHAN_i"    "rLAN_i"    "rMaxPos"   "rMaxPos_i"

## Check 'avi_codebook' in environment to check how each AVI variable was calculated.
```

```r
glimpse(avi_codebook)
```

```
## Rows: 12
## Columns: 2
## $ variables <chr> "rHAP", "rLAP", "rHAN", "rLAN", "rHAP_i", "rLAP_i", "rHAN_i"~
## $ items     <chr> "r.enthusiastic, r.excited, r.elated", "r.relaxed, r.calm, r~
```

```r
# actual affect
data <- ipsatize_avi(data,
                     item_stem = "i.",
                     remove = c("euphoric", "contemptuous", "angry"),
                     maximizing_pos = TRUE)
```

```
## Items ipsatized successfully. Variables created:

##  [1] "iMean"            "iSd"               "i.enthusiastic_i" "i.astonished_i"
##  [5] "i.nervous_i"      "i.dull_i"          "i.quiet_i"        "i.relaxed_i"
##  [9] "i.excited_i"      "i.surprised_i"     "i.elated_i"       "i.sleepy_i"
## [13] "i.still_i"        "i.lonely_i"        "i.strong_i"       "i.passive_i"
## [17] "i.content_i"      "i.sluggish_i"      "i.inactive_i"     "i.sad_i"
## [21] "i.fearful_i"      "i.happy_i"         "i.idle_i"         "i.calm_i"
## [25] "i.unhappy_i"      "i.aroused_i"       "i.hostile_i"      "i.satisfied_i"
## [29] "i.rested_i"       "i.peaceful_i"      "i.serene_i"       "i.noemotion_i"
## [33] "i.energetic_i"    "i.ashamed_i"       "i.disgusted_i"    "i.stressed_i"
## [37] "i.guilty_i"       "i.fatigued_i"      "i.euphoric_i"     "i.contemptuous_i"
## [41] "i.angry_i"

## AVI Variables Computed:
```

```
##  [1] "iHAP"      "iLAP"      "iHAN"      "iLAN"      "iHAP_i"    "iLAP_i"
##  [7] "iHAN_i"    "iLAN_i"    "iMaxPos"   "iMaxPos_i"

## Check 'avi_codebook' in environment to check how each AVI variable was calculated.
```

```r
glimpse(avi_codebook)
```

```
## Rows: 12
## Columns: 2
## $ variables <chr> "iHAP", "iLAP", "iHAN", "iLAN", "iHAP_i", "iLAP_i", "iHAN_i"~
## $ items     <chr> "i.enthusiastic, i.excited, i.elated", "i.relaxed, i.calm, i~
```

```r
# calculate cronbach alpha for HAP and LAP

reliab_avi <- c("r.enthusiastic, r.excited, r.elated",
                "r.relaxed, r.calm, r.peaceful, r.serene",
                "i.enthusiastic, i.excited, i.elated",
                "i.relaxed, i.calm, i.peaceful, i.serene")

reliab_avi_names <- c("rHAP", "rLAP", "iHAP", "iLAP")

all_alphas(data, items = reliab_avi, constructs = reliab_avi_names,
           sensitivity = sensitivity_analyses)
```

```
## # A tibble: 12 x 3
##    construct    `!is.na(subID)` `exclude_sd != 1`
##    <chr>                  <dbl>             <dbl>
##  1 rHAP_overall            0.77              0.76
##  2 rLAP_overall            0.85              0.84
##  3 iHAP_overall            0.63              0.6
##  4 iLAP_overall            0.82              0.79
##  5 rHAP_EA                 0.79              0.77
##  6 rLAP_EA                 0.85              0.83
##  7 iHAP_EA                 0.56              0.51
##  8 iLAP_EA                 0.83              0.77
##  9 rHAP_TN                 0.78              0.76
## 10 rLAP_TN                 0.85              0.84
## 11 iHAP_TN                 0.74              0.72
## 12 iLAP_TN                 0.81              0.82
```

Ipsatized giving scores

Now, we compute the ipsatized scores of the trials. Alike the AVI variables, we calculate this by converting the amount of money given using this formula: `ipsatized score = (raw amoung given - individuals' mean amount given across all trials) / (individuals' sd across the amount of money given across all trials)`

```
# compute ipsatized giving scores
data <- data %>%
  rowwise() %>%
  dplyr::mutate( # create intermediate variables for calculating ipsatize scores
    givingMean = mean(
      c_across(matches("^raw[1-9][1-9]*[0]*$")), na.rm = TRUE),
    givingSd = sd(
      c_across(matches("^raw[1-9][1-9]*[0]*$")), na.rm = TRUE)) %>%
  dplyr::mutate( # create ipsatized item scores
    across(matches("^raw[1-9][1-9]*[0]*$"),
           ~ as.numeric((.x - givingMean) / givingSd),
           .names = "{.col}_i")) %>%
  ungroup()
```

Ipsatized trait rating scores

Finally, we compute the ipsatized scores of the trait ratings for the faces (items that start with `trait`, `vale`, and `arou`).

All participants rated the faces on numerous traits. Specifically, they responded to the prompt "To what extent is this person…" (e.g., "Competent", "Trustworthy", etc.; 1 = *Not at all*, 4 = *Moderately*, 7 = *Extremely*) and how positive ("*How negative - positive do you feel when you see this face?*"; 1 = *Extremely negative*, 7 = *Extremely positive*) and aroused ("*How not-aroused - aroused do you feel when you see this face? Aroused means activated, charged, alert, or energized physically and mentally.*"; 1 = *Not at all aroused*, 7 = *Extremely aroused*) they feel when they see the face. For the repeated ratings, we only include the first rating they provide.

Alike the AVI and giving variables, we calculate this by converting the ratings given using this formula: `ipsatized score = (raw rating - individuals' mean rating across all traits and faces) / (individuals' sd rating across all traits and faces)`. Note that for valence (`vale`) and arousal (`arou`) ratings, they are ipsatized within the valence and arousal ratings only (i.e., `ipsatized valence = (raw valence rating - individuals' mean rating of valence across all faces) / (individuals' sd of valence across all faces)`.

```
# compute ipsatized trait rating scores
data <- data %>%
  rowwise() %>%
  dplyr::mutate( # create intermediate variables for calculating ipsatize scores
    traitRatingMean = mean(
      c_across(matches("^trait[1-9][0-9]*_")), na.rm = TRUE),
    traitRatingSd = sd(
      c_across(matches("^trait[1-9][0-9]*_")), na.rm = TRUE),
    valeRatingMean = mean(
      c_across(matches("^vale[1-9][0-9]*$")), na.rm = TRUE),
    valeRatingSd = sd(
      c_across(matches("^vale[1-9][0-9]*$")), na.rm = TRUE),
    arouRatingMean = mean(
      c_across(matches("^arou[1-9][0-9]*$")), na.rm = TRUE),
    arouRatingSd = sd(
      c_across(matches("^arou[1-9][0-9]*$")), na.rm = TRUE)) %>%
  dplyr::mutate(
    # create ipsatized item scores
    across(matches("^trait[1-9][0-9]*_"),
           ~ as.numeric((.x - traitRatingMean) / traitRatingSd),
           .names = "{.col}_i")) %>%
  dplyr::mutate( # create ipsatized item scores
    across(matches("^vale[1-9][0-9]*$"),
           ~ as.numeric((.x - valeRatingMean) / valeRatingSd),
           .names = "{.col}_i")) %>%
  dplyr::mutate( # create ipsatized item scores
    across(matches("^arou[1-9][0-9]*$"),
           ~ as.numeric((.x - arouRatingMean) / arouRatingSd),
           .names = "{.col}_i")) %>%
  ungroup()
```

Save data computed frame

```
write.csv(data, "ia_study2_anon_data_computed.csv")
```

Descriptives

```
# descriptives function
describe_all <- function(data, filter_condition1) {
  for (i in 1:length(filter_condition1)) {
      subset_data <- data %>%
        filter(eval(parse(text = filter_condition1[i])))
   desc <- subset_data  %>% rowwise() %>%
    dplyr::mutate(
      avg_giving_amt = mean(
        c_across((starts_with("raw") & !contains("_Page.Submit")))),
      avg_giving_RT = mean(
        c_across((starts_with("raw") & contains("_Page.Submit"))))) %>%
      dplyr::select(gender_nonfemale, sexualpref_nonhetero, ethn_nonmajority,
                    age, famises2_numbered, selfses2_numbered,
                    tipi_ex, tipi_a, tipi_c, tipi_es, tipi_o,
                    iri_pt, iri_fs, iri_ec, iri_pd,
                    avg_giving_amt, avg_giving_RT,
                    rHAP, rLAP, iHAP, iLAP) %>%
     psych::describe(fast = TRUE, skew = FALSE)
  print(filter_condition1[i])
  print(desc)
  }
}
```

```r
# Pooled data
describe_all(data = data, sensitivity_analyses)
```

```
## [1] "!is.na(subID)"
##                      vars   n  mean   sd median   min   max range   se
## gender_nonfemale        1 222  0.28 0.45   0.00  0.00  1.00  1.00 0.03
## sexualpref_nonhetero    2 222  0.28 0.45   0.00  0.00  1.00  1.00 0.03
## ethn_nonmajority        3 222  0.06 0.24   0.00  0.00  1.00  1.00 0.02
## age                     4 222 23.56 2.95  23.00 18.00 31.00 13.00 0.20
## famises2_numbered       5 219  5.27 2.08   5.00  1.00  8.00  7.00 0.14
## selfses2_numbered       6 219  2.46 1.91   2.00  1.00  8.00  7.00 0.13
## tipi_ex                 7 221  3.64 1.59   3.50  1.00  7.00  6.00 0.11
## tipi_a                  8 221  4.93 1.18   5.00  1.00  7.00  6.00 0.08
## tipi_c                  9 221  5.02 1.30   5.00  1.00  7.00  6.00 0.09
## tipi_es                10 221  3.86 1.31   4.00  1.00  7.00  6.00 0.09
## tipi_o                 11 221  4.90 1.18   5.00  1.50  7.00  5.50 0.08
## iri_pt                 12 222  3.57 0.65   3.57  2.00  5.00  3.00 0.04
## iri_fs                 13 222  3.50 0.72   3.50  1.14  5.00  3.86 0.05
## iri_ec                 14 222  3.61 0.71   3.57  1.00  5.00  4.00 0.05
## iri_pd                 15 222  3.00 0.71   3.00  1.00  4.57  3.57 0.05
## avg_giving_amt         16 187  0.78 0.55   0.74  0.01  2.76  2.75 0.04
## avg_giving_RT          17 222  5.21 3.03   4.50  1.07 21.91 20.85 0.20
## rHAP                   18 222  2.69 0.80   2.67  1.00  4.67  3.67 0.05
## rLAP                   19 222  2.91 0.84   3.00  1.00  5.00  4.00 0.06
## iHAP                   20 222  3.36 0.75   3.33  1.00  5.00  4.00 0.05
## iLAP                   21 222  3.93 0.76   4.00  1.00  5.00  4.00 0.05
## [1] "exclude_sd != 1"
##                      vars   n  mean   sd median   min   max range   se
## gender_nonfemale        1 187  0.27 0.45   0.00  0.00  1.00  1.00 0.03
## sexualpref_nonhetero    2 187  0.27 0.44   0.00  0.00  1.00  1.00 0.03
## ethn_nonmajority        3 187  0.06 0.25   0.00  0.00  1.00  1.00 0.02
## age                     4 187 23.48 2.84  23.00 18.00 31.00 13.00 0.21
## famises2_numbered       5 186  5.23 2.03   5.00  1.00  8.00  7.00 0.15
## selfses2_numbered       6 185  2.35 1.83   1.00  1.00  8.00  7.00 0.13
## tipi_ex                 7 186  3.66 1.59   3.50  1.00  7.00  6.00 0.12
## tipi_a                  8 186  4.94 1.20   5.00  1.00  7.00  6.00 0.09
## tipi_c                  9 186  4.98 1.30   5.00  2.00  7.00  5.00 0.09
## tipi_es                10 186  3.84 1.27   4.00  1.00  7.00  6.00 0.09
## tipi_o                 11 186  4.90 1.12   4.75  2.00  7.00  5.00 0.08
## iri_pt                 12 187  3.58 0.65   3.57  2.00  5.00  3.00 0.05
## iri_fs                 13 187  3.52 0.71   3.43  1.71  5.00  3.29 0.05
## iri_ec                 14 187  3.58 0.72   3.57  1.00  5.00  4.00 0.05
## iri_pd                 15 187  3.03 0.69   3.14  1.00  4.57  3.57 0.05
## avg_giving_amt         16 187  0.78 0.55   0.74  0.01  2.76  2.75 0.04
## avg_giving_RT          17 187  5.50 2.99   4.77  1.91 21.91 20.01 0.22
## rHAP                   18 187  2.72 0.80   2.67  1.00  4.67  3.67 0.06
## rLAP                   19 187  2.95 0.82   3.00  1.00  5.00  4.00 0.06
## iHAP                   20 187  3.36 0.75   3.33  1.00  5.00  4.00 0.05
## iLAP                   21 187  3.94 0.73   4.00  1.33  5.00  3.67 0.05
```

```r
# European American
describe_all(data = data %>% dplyr::filter(cultureEA == 1), sensitivity_analyses)
```

```
## [1] "!is.na(subID)"
##                       vars   n  mean   sd median   min   max range   se
## gender_nonfemale         1 116  0.29 0.46   0.00  0.00  1.00  1.00 0.04
## sexualpref_nonhetero     2 116  0.31 0.46   0.00  0.00  1.00  1.00 0.04
## ethn_nonmajority         3 116  0.06 0.24   0.00  0.00  1.00  1.00 0.02
## age                      4 116 23.93 3.09  24.00 18.00 31.00 13.00 0.29
## famises2_numbered        5 114  6.46 1.69   7.00  1.00  8.00  7.00 0.16
## selfses2_numbered        6 115  3.30 2.18   3.00  1.00  8.00  7.00 0.20
## tipi_ex                  7 115  3.61 1.61   3.50  1.00  7.00  6.00 0.15
## tipi_a                   8 115  4.98 1.17   5.00  1.50  7.00  5.50 0.11
## tipi_c                   9 115  5.43 1.22   6.00  2.50  7.00  4.50 0.11
## tipi_es                 10 115  3.93 1.33   4.00  1.00  7.00  6.00 0.12
## tipi_o                  11 115  5.01 1.21   5.00  2.00  7.00  5.00 0.11
## iri_pt                  12 116  3.53 0.69   3.54  2.00  5.00  3.00 0.06
## iri_fs                  13 116  3.54 0.74   3.57  1.14  4.75  3.61 0.07
## iri_ec                  14 116  3.80 0.64   3.86  2.43  5.00  2.57 0.06
## iri_pd                  15 116  2.88 0.69   3.00  1.00  4.33  3.33 0.06
## avg_giving_amt          16  91  0.82 0.59   0.82  0.01  2.76  2.75 0.06
## avg_giving_RT           17 116  5.65 3.40   4.74  1.52 21.91 20.40 0.32
## rHAP                    18 116  2.72 0.85   2.67  1.00  4.67  3.67 0.08
## rLAP                    19 116  2.76 0.88   2.75  1.00  5.00  4.00 0.08
## iHAP                    20 116  3.50 0.76   3.67  1.00  5.00  4.00 0.07
## iLAP                    21 116  3.95 0.81   4.00  1.00  5.00  4.00 0.08
## [1] "exclude_sd != 1"
##                       vars  n  mean   sd median   min   max range   se
## gender_nonfemale         1 91  0.31 0.46   0.00  0.00  1.00  1.00 0.05
## sexualpref_nonhetero     2 91  0.26 0.44   0.00  0.00  1.00  1.00 0.05
## ethn_nonmajority         3 91  0.07 0.25   0.00  0.00  1.00  1.00 0.03
## age                      4 91 23.78 2.93  24.00 18.00 31.00 13.00 0.31
## famises2_numbered        5 90  6.42 1.66   7.00  1.00  8.00  7.00 0.17
## selfses2_numbered        6 90  3.20 2.14   3.00  1.00  8.00  7.00 0.23
## tipi_ex                  7 90  3.63 1.58   4.00  1.00  7.00  6.00 0.17
## tipi_a                   8 90  5.02 1.17   5.00  2.50  7.00  4.50 0.12
## tipi_c                   9 90  5.37 1.29   6.00  2.50  7.00  4.50 0.14
## tipi_es                 10 90  3.91 1.29   4.00  1.00  7.00  6.00 0.14
## tipi_o                  11 90  5.02 1.18   5.00  2.50  7.00  4.50 0.12
## iri_pt                  12 91  3.53 0.71   3.57  2.00  5.00  3.00 0.07
## iri_fs                  13 91  3.58 0.71   3.43  1.86  4.75  2.89 0.07
## iri_ec                  14 91  3.77 0.68   3.71  2.43  5.00  2.57 0.07
## iri_pd                  15 91  2.95 0.67   3.14  1.00  4.33  3.33 0.07
## avg_giving_amt          16 91  0.82 0.59   0.82  0.01  2.76  2.75 0.06
## avg_giving_RT           17 91  6.09 3.34   5.22  2.23 21.91 19.68 0.35
## rHAP                    18 91  2.80 0.88   2.67  1.00  4.67  3.67 0.09
## rLAP                    19 91  2.81 0.86   2.75  1.00  5.00  4.00 0.09
## iHAP                    20 91  3.53 0.77   3.67  1.00  5.00  4.00 0.08
## iLAP                    21 91  3.97 0.76   4.00  1.33  5.00  3.67 0.08
```

```r
# Taiwanese
describe_all(data = data %>% dplyr::filter(cultureEA == 0), sensitivity_analyses)
```

```
## [1] "!is.na(subID)"
##                      vars   n  mean   sd median   min   max range   se
## gender_nonfemale        1 106  0.26 0.44   0.00  0.00  1.00  1.00 0.04
## sexualpref_nonhetero    2 106  0.25 0.44   0.00  0.00  1.00  1.00 0.04
## ethn_nonmajority        3 106  0.06 0.23   0.00  0.00  1.00  1.00 0.02
## age                     4 106 23.16 2.75  23.00 18.00 30.00 12.00 0.27
## famises2_numbered       5 105  3.99 1.67   4.00  1.00  8.00  7.00 0.16
## selfses2_numbered       6 104  1.52 0.90   1.00  1.00  5.00  4.00 0.09
## tipi_ex                 7 106  3.67 1.57   3.50  1.00  6.50  5.50 0.15
## tipi_a                  8 106  4.87 1.20   5.00  1.00  7.00  6.00 0.12
## tipi_c                  9 106  4.58 1.24   4.50  1.00  7.00  6.00 0.12
## tipi_es                10 106  3.78 1.28   4.00  1.00  6.50  5.50 0.12
## tipi_o                 11 106  4.78 1.13   4.50  1.50  7.00  5.50 0.11
## iri_pt                 12 106  3.61 0.61   3.57  2.00  5.00  3.00 0.06
## iri_fs                 13 106  3.46 0.71   3.43  1.71  5.00  3.29 0.07
## iri_ec                 14 106  3.40 0.72   3.43  1.00  4.71  3.71 0.07
## iri_pd                 15 106  3.13 0.71   3.07  1.00  4.57  3.57 0.07
## avg_giving_amt         16  96  0.74 0.51   0.64  0.01  2.54  2.53 0.05
## avg_giving_RT          17 106  4.72 2.50   4.35  1.07 15.40 14.33 0.24
## rHAP                   18 106  2.66 0.74   2.67  1.00  4.33  3.33 0.07
## rLAP                   19 106  3.08 0.77   3.00  1.25  5.00  3.75 0.07
## iHAP                   20 106  3.20 0.71   3.00  1.67  5.00  3.33 0.07
## iLAP                   21 106  3.92 0.71   4.00  2.25  5.00  2.75 0.07
## [1] "exclude_sd != 1"
##                      vars  n  mean   sd median   min   max range   se
## gender_nonfemale        1 96  0.24 0.43   0.00  0.00  1.00  1.00 0.04
## sexualpref_nonhetero    2 96  0.27 0.45   0.00  0.00  1.00  1.00 0.05
## ethn_nonmajority        3 96  0.06 0.24   0.00  0.00  1.00  1.00 0.02
## age                     4 96 23.20 2.73  23.00 18.00 30.00 12.00 0.28
## famises2_numbered       5 96  4.10 1.68   4.00  1.00  8.00  7.00 0.17
## selfses2_numbered       6 95  1.55 0.93   1.00  1.00  5.00  4.00 0.10
## tipi_ex                 7 96  3.69 1.60   3.50  1.00  6.50  5.50 0.16
## tipi_a                  8 96  4.88 1.22   5.00  1.00  7.00  6.00 0.12
## tipi_c                  9 96  4.61 1.20   4.50  2.00  7.00  5.00 0.12
## tipi_es                10 96  3.78 1.27   4.00  1.00  6.50  5.50 0.13
## tipi_o                 11 96  4.79 1.05   4.50  2.00  7.00  5.00 0.11
## iri_pt                 12 96  3.63 0.58   3.71  2.00  5.00  3.00 0.06
## iri_fs                 13 96  3.46 0.71   3.43  1.71  5.00  3.29 0.07
## iri_ec                 14 96  3.40 0.73   3.43  1.00  4.71  3.71 0.07
## iri_pd                 15 96  3.11 0.71   3.00  1.00  4.57  3.57 0.07
## avg_giving_amt         16 96  0.74 0.51   0.64  0.01  2.54  2.53 0.05
## avg_giving_RT          17 96  4.94 2.51   4.57  1.91 15.40 13.49 0.26
## rHAP                   18 96  2.64 0.71   2.67  1.33  4.33  3.00 0.07
## rLAP                   19 96  3.08 0.75   3.00  1.25  4.75  3.50 0.08
## iHAP                   20 96  3.19 0.70   3.00  1.67  5.00  3.33 0.07
## iLAP                   21 96  3.92 0.71   4.00  2.25  5.00  2.75 0.07
```